Experimental Protocol: Tannic-Acid-Functionalized Fish-Scale Ossein Biosorbent for Selective Lead(II) Capture and Recovery from Multi-Metal Water

Date	8 June 2026
Material	Demineralized fish-scale ossein, functionalized with tannic acid (TA–OSS); unfunctionalized ossein (RAW–OSS) as control
Analytes	Lead(II), copper(II), zinc(II)
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Contents

1 Purpose and scope

This protocol is a complete, self-contained procedure for preparing a tannic-acid-functionalized fish-scale ossein biosorbent (TA–OSS) and measuring its performance for the selective capture and recovery of lead(II) from water that also contains competing copper(II) and zinc(II). It is written to be followed directly at the bench: read it in order and carry out each step.

The work runs in three phases. **Phase A** conditions the supplied ossein to a granular fraction, functionalizes it with tannic acid, and characterizes the product. **Phase B** measures batch equilibrium capacity, adsorption kinetics, competitive selectivity in lead/copper/zinc mixtures, and regeneration over repeated cycles. **Phase C** packs the sorbent into a fixed-bed column and measures breakthrough behaviour, counter-current acid regeneration, lead enrichment in the regenerate, and capacity retention across service–regeneration cycles. Only lead(II), copper(II), and zinc(II) are studied.

Note. Throughout, all dissolving, mixing, and equilibration are carried out on a magnetic stirrer, and all metal analysis is by flame atomic absorption spectrometry (AAS). The target a successful measurement should meet is flagged with an **Expected** note in each procedure.

2 Definitions and abbreviations

Term	Definition
Ossein	Demineralized collagen matrix remaining after the mineral (hydroxyapatite) phase is removed from fish scales. Supplied here in bulk.
TA–OSS	Tannic-acid-functionalized ossein: the material under test.
RAW–OSS	Unfunctionalized ossein processed identically but without tannic acid: the negative control.
Galloyl group	3,4,5-trihydroxybenzoyl (vicinal trihydroxy aromatic) unit of tannic acid; the oxygen-donor chelation site responsible for lead selectivity.
q_e, $q_{\max}$	Equilibrium adsorption capacity and Langmuir maximum monolayer capacity ($\mathrm{mg\,g^{-1}}$).
α(Pb/Cu), α(Pb/Zn)	Selectivity coefficient of lead over the competing metal; a value above 1 means preferential lead uptake.
K_d	Distribution coefficient, q_e/C_e ($\mathrm{L\,g^{-1}}$).
BV	Bed volume: the measured volume of the packed sorbent bed (mL). One BV of throughput equals one packed-bed volume of liquid.
BV_{10}, BV_{50}, BV_{90}	Bed volumes of feed processed when the effluent reaches 10%, 50%, and 90% of the feed concentration.
EBCT	Empty-bed contact time, $V_{\text{bed}}/Q = 1/(\text{BV per hour})$.
EF	Enrichment factor: lead concentration in the regenerate divided by lead concentration in the feed.
pH_{PZC}	pH of the point of zero charge of the sorbent surface.
AAS	Flame atomic absorption spectrometry.

3 Health, safety, and environment

- **Acids.** Concentrated HCl and HNO_3 are corrosive and give off irritating vapour. Prepare and dispense all acid solutions in a fume hood. Always add acid to water, never water to acid. Wear nitrile gloves, a lab coat, and goggles; use acid-resistant gloves during column regeneration with 0.1 M HCl.
- **Sodium hydroxide.** Pellets and solutions are caustic; avoid skin and eye contact.
- **Lead salts.** $Pb(NO_3)_2$ and all lead-bearing solutions, filtrates, spent sorbent, and acid regenerates are toxic. Collect them as heavy-metal hazardous waste; do not pour to drain. Label every lead-bearing stream.
- **Tannic acid and ossein.** Low hazard; wear a dust mask when grinding to avoid inhaling fines.
- **Hot surfaces.** The drying oven (50–60 °C), muffle furnace (550–600 °C), and stirrer hotplate are burn hazards; use tongs and heat-resistant gloves.
- **Glassware under vacuum.** Use only undamaged borosilicate for Buchner filtration.
- **Waste segregation.** Keep lead-bearing aqueous waste, spent sorbent, and acid regenerate in separate labelled containers.

4 Reagents and materials

All reagents are ACS reagent grade or equivalent. Quantities are sized for the full programme (material preparation, batch testing, and two packed columns) with a working margin.

Reagent	Specification	Qty	Purpose
Tannic acid ($C_{76}H_{52}O_{46}$)	ACS, $\geq$ 95%	25 g	Functionalization ligand (galloyl O-donor sites)
Lead nitrate, $Pb(NO_3)_2$	ACS grade	25 g	Lead stock for all phases and the column feed
Copper sulfate pentahydrate, $CuSO_4{\cdot}5H_2O$	ACS grade	15 g	Copper stock (competitor)
Zinc sulfate heptahydrate, $ZnSO_4{\cdot}7H_2O$	ACS grade	15 g	Zinc stock (competitor)
Hydrochloric acid, HCl	37% conc.	1 L	pH adjustment, batch desorption, column regeneration (0.1 M)
Sodium hydroxide, NaOH	Pellets, ACS	100 g	pH adjustment; functionalization bath at pH 5.0
Nitric acid, HNO_3	1 M	250 mL	Stock preservation and 5% glassware acid wash
Sodium chloride, NaCl	ACS grade	10 g	0.01 M background electrolyte for pH_{PZC} only
Deionized / distilled water	—	30 L	Solution preparation, washing, column rinsing
pH buffer standards	pH 4, 7, 10	1 set	Three-point pH-meter calibration
Glass wool	Acid-washed borosilicate	1 pack	Column bed support plug and top cap

Note. The starting material is demineralized fish-scale ossein supplied in bulk (approximately

25 g). It is conditioned to a granular working fraction in Section 7.2. No demineralization is performed in this protocol.

5 Apparatus and equipment

5.1 Analytical and preparative instruments

Instrument	Specification	Use
Flame AAS	Hollow-cathode lamps for Pb (283.3 nm), Cu (324.8 nm), Zn (213.9 nm); air–acetylene flame	All metal analysis (one element per aspiration)
ATR-FTIR spectrometer	ATR accessory; 4000–400 cm^{-1}, 4 cm^{-1} resolution, $\geq$ 32 scans	Characterization: RAW–OSS, TA–OSS, and spent TA–OSS
Multi-position magnetic stirrer	Several stir positions; PTFE stir bars; moderate speed	All batch contacting and equilibration in parallel
Magnetic stirrer with hotplate	Variable speed; gentle heating to 40 °C	Reagent and stock dissolution; tannic-acid functionalization
Analytical balance	0.0001 g readability	Sorbent dosing, salt and tannic-acid weighing, gravimetry
pH meter and electrode	Three-point calibration (pH 4, 7, 10)	All pH measurement and adjustment
Drying oven	Adjustable 50–60 °C	Ossein and sorbent drying to constant mass
Muffle furnace, crucibles, desiccator	Furnace to 550–600 °C; tared crucibles	Residual-mineral (ash) content
Vacuum (Buchner) filtration	Buchner funnel with filter paper or sintered funnel	Bulk washing of functionalized sorbent
Mortar and pestle; sieves	0.5 mm and 1.0 mm mesh	Grinding and sizing of ossein
UV-Vis spectrophotometer (optional)	Quartz cuvette, 276 nm	Quantitative wash endpoint and tannic-acid leaching check

5.2 Fixed-bed column hardware

The committed column dimensions are derived in Section 5.4.

Component	Specification	Notes
Glass chromatography column	Borosilicate; internal diameter 10 mm; length 250 mm; PTFE frit at base; adjustable PTFE top adapter	Two units (Column A for single-metal cycles, Column B for the ternary run), or one unit cleaned and re-packed
Peristaltic pump	Variable 0.1–10 $mL\,min^{-1}$; two channels preferred	Feed, regenerant, and rinse; a programmable pump enables unattended long runs

Component	Specification	Notes
Tubing	PTFE on the 0.1 M HCl line; Tygon or Norprene (1.6 mm ID) for feed and rinse	Use PTFE only on the acid line; flexible PVC degrades in dilute HCl
In-line pH electrode (recommended)	Outlet flow-cell electrode; pH-strip fallback	Confirms service-pH stability and rinse completion
Fraction collector (optional) and vials	Auto-sampler; $\geq 120 \times 20$ mL vials	Unattended fraction collection; manual collection acceptable if staffed
Feed reservoir, stand, clamps	1 L cylinder or bottle; ring stand	Holds the column vertical; reservoir feeds the pump
Acid-resistant gloves and goggles	Nitrile and impact goggles	Required during HCl regeneration

5.3 Glassware and consumables

Item	Qty	Notes
Volumetric flasks, 1 L	4	Lead, copper, zinc stocks plus one spare
Volumetric flasks, 500 mL	4	Working solutions and column feed batches
Volumetric flasks, 100 mL	15	Calibration standards and dilutions
Erlenmeyer (conical) flasks, 250 mL	≥ 40	All batch contacting (each on a stir position)
Beakers, 100–1000 mL	≈ 20	Functionalization bath and solution prep
Graduated cylinders, 10 / 50 / 100 mL	2 each	Volume measurement
Syringe filters 0.45 μm and syringes, 10 mL	≈ 150 each	Post-adsorption and fraction filtration
Petri dishes	6	Sorbent drying
Micropipette 100–1000 μL and tips	1 set	Precise dispensing
Sample / fraction vials, 20–50 mL	≈ 150	Sample, sorbent, and column-fraction storage
Parafilm; wash bottles	1 roll; 2	Flask covering; DI rinsing

5.4 Column sizing and design basis

The column is sized from first principles: a meaningful breakthrough experiment, suppressed wall effects, a workable contact time, and economy of the limited sorbent. Each consideration is stated, the sizing is calculated, and a single design is committed.

5.4.1 Constraints

- **Wall effect.** To suppress wall channeling, the column-to-particle diameter ratio D_c/d_p should be at least about 10. The column-packing fraction is set to 0.50–1.00 mm, so $d_p \leq 1.0$ mm, and a 10 mm internal diameter gives $D_c/d_p \approx 10$–20.

- **Contact time.** The empty-bed contact time, EBCT = 1/(BV per hour), must be long enough for galloyl chelation. At 8 BV h^{-1} the EBCT is about 7.5 min, ample for the fast lead ligand exchange this system relies on.
- **Material budget.** About 25 g of ossein is available. Batch testing and characterization consume the majority, so the column bed is kept to a few grams while remaining long enough for a high aspect ratio.
- **Analytical and time budget.** A smaller bed reduces the absolute feed volume and the number of fractions to analyze for a given number of bed volumes.

5.4.2 Sizing calculation and committed dimensions

A 10 mm internal diameter and a charge of 4.0 g of TA–OSS are selected. With a packed-bed density of $\rho_b \approx 0.45$–0.55 g cm^{-3}, the bed volume is $V_{\text{bed}} = m/\rho_b \approx 7.3$–$8.9$ mL. The cross-sectional area is $A = \pi(0.5\text{ cm})^2 = 0.785\text{ cm}^2$, so the bed height is $V_{\text{bed}}/A \approx 9.3$–$11.3$ cm and the aspect ratio is about 9:1 to 11:1. One bed volume (BV) is the measured packed-bed volume, confirmed at packing (Section 9.1).

Parameter	Committed value	Basis
Column body / material	Borosilicate, ID 10 mm, length 250 mm	Corrosion-resistant; length fits the bed plus freeboard and adapter
Sorbent charge	4.0 g TA–OSS	Economical on supplied ossein; gives a high-aspect-ratio bed
Particle fraction	0.50–1.00 mm	Sieve-defined; gives $D_c/d_p \approx 10$–20 and limits channeling
Packed-bed density ρ_b	0.45–0.55 g cm^{-3} (measured)	Typical for this granular ossein; measured at packing
Bed volume BV	$\approx$ 7.3–8.9 mL (measured)	$= m/\rho_b$
Bed height	$\approx$ 9.3–11.3 cm (measured)	$= V_{\text{bed}}/A$
Aspect ratio H:D	$\approx$ 9:1 to 11:1 (measured)	High aspect ratio for a sharp front
D_c/d_p	$\approx$ 10–20	$\geq$ 10 suppresses wall effects
Service feed (Column A)	100 mg L^{-1} lead	Tractable breakthrough and strong enrichment (below)
Service pH	5.0	Optimum confirmed in Section 8.1
Service flow / direction	8 BV h^{-1} downflow ($\approx$ 1.0 mL min^{-1})	EBCT $\approx$ 7.5 min
Regeneration eluent	0.1 M HCl	Dilute mineral acid; concentrates the desorbed lead
Regeneration flow / volume	$\approx$ 3 BV at 4 BV h^{-1} upflow ($\approx$ 45 min)	Counter-current for a sharp, concentrated regenerate
Rinse	DI water (pH $\approx$ 5) downflow at 8 BV h^{-1}	Until outlet pH within $\pm$0.3 of inlet
Cycles	3 on Column A; 1 ternary run on Column B	Capacity retention and multi-metal handling

5.4.3 Expected breakthrough and feed choice

For a fixed bed fed at concentration C_0, the bed volumes to stoichiometric saturation are approximately $\mathrm{BV_{sat}} \approx q^* \rho_b / C_0$, where q^* is the equilibrium loading at C_0. Using a representative monolayer capacity of order 40 $\mathrm{mg\,g^{-1}}$ (the value this system is expected to deliver, confirmed in Phase B) with an affinity constant of order 0.05 $\mathrm{L\,mg^{-1}}$ and $\rho_b \approx 0.5$ $\mathrm{g\,cm^{-3}}$, the planning estimates are below. They are **estimates only**; the breakthrough bed volume is a measured output, and because dynamic capacity is typically 60–90% of equilibrium capacity the measured $\mathrm{BV_{10}}$ is usually lower than the saturation figure.

Feed ($\mathrm{mg\,L^{-1}}$)	q^*	$\mathrm{BV_{sat}}$	**$\mathrm{BV_{10}}$ est.**	**Time at 8 BV/h**	**EF (3 BV)**
50	≈ 29	≈ 286	≈ 200	≈ 25 h	$\approx 95\times$
100 (selected)	≈ 33	≈ 167	≈ 117	≈ 15 h	$\approx 56\times$
200 (accelerated)	≈ 36	≈ 91	≈ 64	≈ 8 h	$\approx 30\times$
300	≈ 38	≈ 63	≈ 44	≈ 5.5 h	$\approx 21\times$

A feed of 100 $\mathrm{mg\,L^{-1}}$ is selected for Column A: the breakthrough is tractable with a programmable pump and the regenerate enrichment is of order 50×.

Caution. A high-capacity sorbent at a dilute feed processes many bed volumes before breakthrough, so a service run is long (many hours). These accommodations are authorized and do not compromise the data: use a programmable pump and fraction collector; pause the pump (standby) overnight with the bed kept wetted; terminate cycles 2 and 3 once the effluent reaches $C/C_0 \approx 0.30$ (the $\mathrm{BV_{10}}$ retention metric is captured before that point); and, if needed, use a 200 $\mathrm{mg\,L^{-1}}$ feed and/or a higher velocity (up to about 15 $\mathrm{BV\,h^{-1}}$) to roughly halve the run time, at the cost of a broader mass-transfer zone.

6 Global parameters

Unless a step states otherwise, the following apply.

Parameter	**Value**
Sorbent dose (batch)	0.1 g in 50 mL (2 $\mathrm{g\,L^{-1}}$)
Contacting	Magnetic stirrer, moderate speed (about 300–400 rpm) for all dissolving, mixing, and equilibration
Default contact time (batch)	120 min (validate against the kinetic curve, Section 8.3; shorten later if equilibrium is reached earlier)
Temperature	Ambient (≈ 25 °C)
Target metals	Lead, copper, zinc
Analysis	Flame AAS (one element per aspiration)
Filtration	0.45 μm syringe filter
Replication	$n = 3$ on headline points (50 ppm lead isotherm point; equimolar ternary; every regeneration cycle); $n = 2$ minimum elsewhere; report mean ± SD
Operating pH	From Section 8.1 (expected 4.5–5.5)

Parameter	Value
Column service flow	8 BV h^{-1} downflow (Section 9)
Column regeneration flow	$\approx$ 4 BV h^{-1} counter-current upflow

7 Phase A: sorbent preparation and characterization

7.1 Glassware acid wash and stock solutions

Acid wash. Soak all glassware in about 5% HNO_3 (50 mL of 1 M HNO_3 made to 1 L with DI) for 10–15 min, rinse three times with DI, and air-dry.

Stock solutions (1000 ppm each). Dissolve the mass below in about 800 mL DI in a 1 L volumetric flask on a magnetic stirrer, add 1 mL of 1 M HNO_3, fill to the mark, and invert to mix.

Metal	Salt	Mass for 1 L
Lead	$Pb(NO_3)_2$	1.599 g
Copper	$CuSO_4{\cdot}5H_2O$	3.929 g
Zinc	$ZnSO_4{\cdot}7H_2O$	4.398 g

pH-adjustment solutions. 0.1 M HCl (4.15 mL conc. to 500 mL); 1 M HCl (41.5 mL conc. to 500 mL); 0.1 M NaOH (2.0 g to 500 mL); 1 M NaOH (8.0 g to 200 mL).

Calibration standards. Prepare five standards per metal (1, 5, 10, 25, 50 ppm) by dilution from the 1000 ppm stock, each with 0.1 mL of 1 M HNO_3 per 100 mL to match the sample matrix. Include a blank. Every curve must reach $R^2 > 0.99$. Run a mid-range check standard at the start, middle, and end of each analytical batch.

Note. Complete all calibration before any adsorption sample is run.

7.2 Ossein conditioning

The supplied ossein is demineralized in bulk and not yet powdered.

1. Rinse the ossein three times with 500 mL portions of DI water.
2. Dry in the oven at 60 °C until constant mass ($\Delta m < 0.5\%$ over 1 h). Record the dry mass.
3. Grind with a mortar and pestle.
4. Sieve through the 1.0 mm screen, then the 0.5 mm screen; collect the 0.50–1.00 mm fraction (passes 1.0 mm, retained on 0.5 mm). Record the collected mass.
5. Portion: 18 g for functionalization (Section 7.3), 5 g for the RAW–OSS control plus ash and FTIR samples, and the remainder (about 2 g) as reserve.

Rationale. The 0.50–1.00 mm fraction gives $D_c/d_p \approx 10$–20 at the 10 mm column ID, suppressing channeling while remaining easy to pack and recover.

7.3 Tannic-acid functionalization

TA–OSS. On a magnetic stirrer:

1. Dissolve 18 g tannic acid in 360 mL DI (5% w/v) in a beaker; warm gently to $\leq$ 50 °C if

dissolution is slow.

2. Adjust the bath to pH 5.0 with 1 M NaOH, added dropwise. Record the volume used.
3. Add the 18 g of sized ossein. Cover the beaker with parafilm. Stir for 12–16 h (overnight).

Caution. Keep the stirrer at a moderate speed that circulates the granules without grinding them. Excessive speed pulverizes granules into fines that later foul column packing and bias surface area. If granules settle and stop moving, briefly raise the speed or swirl by hand.

RAW–OSS. Place 5 g of sized ossein in 100 mL DI at pH 5.0; stir identically for the same time. Identical processing without tannic acid isolates the effect of functionalization.

7.4 Washing, drying, loading, portioning

Washing (both batches).

1. Vacuum-filter through a Buchner funnel.
2. Wash with successive 200 mL portions of DI until the filtrate is water-clear (typically 5–8 washes). If a UV-Vis is available, wash until the filtrate absorbance at 276 nm is below 0.05.
3. Wash RAW–OSS with the same number of washes as TA–OSS.

Critical. Insufficient washing leaves free tannic acid that leaches during adsorption, complexes metals in solution, and falsely raises apparent removal. Do not skimp on this step.

Drying and loading.

1. Dry both batches at 50 °C for 6–8 h to constant mass. Record both dry masses.
2. Tannin loading (%) = $(m_{\text{func}} - m_{\text{raw,eq}})/m_{\text{raw,eq}} \times 100$, where m_{func} is the dry mass after functionalization and $m_{\text{raw,eq}}$ the equivalent dry mass of unfunctionalized ossein.

Expected. Tannin loading in the range 4–8 wt%.

Portioning.

1. TA–OSS: at least thirty pre-weighed 0.1 g vials for batch work; one 4.0 g vial for Column A; one 4.0 g vial for Column B; about 1 g for ash; and a small FTIR portion. Store in a desiccator.
2. RAW–OSS: at least twelve pre-weighed 0.1 g vials, plus about 1 g for ash and a small FTIR portion.
3. Photograph RAW–OSS against TA–OSS to document the colour change.

7.5 Characterization

7.5.1 ATR-FTIR (primary)

Record spectra for RAW–OSS, TA–OSS, and (after Section 9.8) the spent TA–OSS. Press the sample onto the ATR crystal; scan 4000–400 cm^{-1} at 4 cm^{-1} resolution with at least 32 scans; export each spectrum and make a labelled overlay.

Expected. Relative to RAW–OSS, TA–OSS shows new aromatic C=C ($\approx$ 1600–1620 cm^{-1}) and galloyl ester C–O / C=O ($\approx$ 1700–1730 cm^{-1}) bands, with the collagen Amide I ($\approx$ 1630 cm^{-1}), Amide II ($\approx$ 1540 cm^{-1}), and Amide III ($\approx$ 1235 cm^{-1}) bands preserved. A phosphate band ($\approx$ 1030 cm^{-1}) indicates residual mineral. The spent sample shows shifts of the carbonyl/ester features and dampening of the phenolic O–H band.

7.5.2 Residual mineral (ash) content

1. Dry about 1 g each of RAW–OSS and TA–OSS to constant mass at 60 °C and weigh into tared crucibles (m_{dry}).

2. Ash in the muffle furnace at 550–600 °C for at least 4 h to constant ash mass. Cool in a desiccator and weigh (m_{ash}).
3. Residual mineral (%) = $(m_{ash}/m_{dry}) \times 100$. Run in duplicate.

Expected. Residual mineral of a few weight percent: the supplied ossein is well-demineralized while retaining a small phosphate fraction, consistent with the $\approx 1030\ cm^{-1}$ FTIR band.

7.5.3 Point of zero charge (supporting)

1. Prepare 0.01 M NaCl. Dispense 50 mL into nine flasks and set their initial pH across 2–10 in one-unit steps. Prepare a parallel set of nine flasks for RAW–OSS.
2. Add 0.1 g of the appropriate sorbent to each. Stir for 24 h on the magnetic stirrer.
3. Measure the final pH of each flask; plot final against initial pH. The pH_{PZC} is the plateau.

Note. A lower pH_{PZC} for TA–OSS than for RAW–OSS confirms enrichment of acidic surface groups from functionalization, and serves as a fallback if the FTIR is inconclusive.

8 Phase B: batch equilibrium and selectivity

All contacting is on the magnetic stirrer at ambient temperature.

8.1 pH optimization

Critical. The optimum pH found here is used in every later batch step and in the column work. Settle it before continuing.

1. Prepare 500 mL of 50 ppm lead from stock.
2. Distribute 50 mL into four flasks set to pH 3.0, 4.0, 5.0, 6.0. Prepare an identical four-flask RAW–OSS set (eight flasks).
3. Add 0.1 g of the appropriate sorbent to each. Stir at moderate speed for 120 min.
4. Filter through 0.45 μm. Measure final pH and residual lead by AAS.
5. Run one sorbent-free 50 ppm lead flask at pH 6.0 as a precipitation control.

Calculate removal (%) = $(C_0 - C_e)/C_0 \times 100$ and $q_e = (C_0 - C_e)V/m$. The pH giving the highest TA–OSS removal is the optimum.

Expected. Optimum in the range pH 4.5–5.5 (likely 5.0). Removal rises from pH 3 to 5 as galloyl groups deprotonate; do not operate above pH 6, where lead hydroxide can precipitate (checked by the sorbent-free control).

8.2 Single-metal isotherms

At the optimum pH, 120 min.

- **Lead.** Six concentrations: 10, 25, 50, 100, 200, 300 ppm, each with 0.1 g TA–OSS in 50 mL. Run the 50 ppm point as $n = 3$. Include one RAW–OSS comparator at 50 ppm.
- **Copper and zinc.** Four concentrations each (10, 50, 100, 300 ppm) with 0.1 g TA–OSS.

Stir 120 min, filter, and measure residual metal by AAS.

Expected. Langmuir maximum capacity for lead in the range 35–70 $mg\,g^{-1}$, with Langmuir fitting better than Freundlich ($R^2 > 0.99$). Single-metal capacity order $q_{max}(Pb) \geq q_{max}(Cu) > q_{max}(Zn)$: lead is not out-competed by copper, the signature of the reversed thermodynamic ordering. Fit the isotherms yourself.

8.3 Adsorption kinetics

Two time-courses, lead and copper, at the optimum pH. **Lead (7 flasks).**

1. Prepare 600 mL of 50 ppm lead at the optimum pH.
2. Distribute 50 mL into six sacrificial flasks. Add 0.1 g TA–OSS to each at staggered start times so all are sampled at the same clock time but represent contact times of 5, 15, 30, 60, 120, 240 min.
3. At each point, immediately filter through 0.45 μm and measure residual lead.
4. Run a duplicate of the 60 min point.

Copper (7 flasks). Identical with 50 ppm copper, same timepoints, same duplicate.

Expected. Lead reaches equilibrium faster and/or shows higher early uptake than copper, supporting the kinetic basis of the selectivity. This also validates the 120 min contact time.

8.4 Competitive (mixed-metal) adsorption

The centrepiece test of selectivity. Two ternary experiments at the optimum pH. **Equimolar ($n = 3$, plus one control).**

1. Prepare 50 mL of 50 ppm lead + 50 ppm copper + 50 ppm zinc.
2. Before adding sorbent, withdraw a 5 mL aliquot and measure the actual initial concentration of all three metals (do not assume nominal values).
3. Add 0.1 g TA–OSS. Stir 120 min. Filter. Measure all three residual metals. Run in triplicate.
4. Run one parallel flask with 0.1 g RAW–OSS as control.

Minority-target ($n = 3$). 50 mL of 25 ppm lead + 100 ppm copper + 100 ppm zinc; measure the initial aliquot, add 0.1 g TA–OSS, stir 120 min, filter, measure all three. Tests whether lead selectivity holds when lead is the minority species (about a 4:1 disadvantage).

Critical. Do not discard the spent sorbent from the equimolar experiment. Store it under DI water in sealed vials for the regeneration runs in Section 8.5.

Calculate for each metal q_e, removal (%), and $K_d = q_e/C_e$. Selectivity $\alpha(\mathrm{Pb/Cu})$ and $\alpha(\mathrm{Pb/Zn})$ from $\alpha(\mathrm{Pb}/M) = (q_{\mathrm{Pb}}\, C_{e,M})/(q_M\, C_{e,\mathrm{Pb}})$.

Note. AAS reads one element at a time, so each ternary sample is aspirated three times (lead, copper, zinc).

Expected. $\alpha(\mathrm{Pb/Cu})$ in the range 3–15 (and above 2.5) and $\alpha(\mathrm{Pb/Zn})$ in the range 5–25, both above 1 under the minority condition. RAW–OSS shows much weaker preference (near unity), confirming that functionalization produces the selectivity.

8.5 Batch regeneration cycling

Material. Use the spent TA–OSS from the equimolar experiment. Run three cycles in triplicate, using three separate 0.1 g aliquots so each cycle is $n = 3$.

Stage A: desorption (30 min). Place 0.1 g of spent sorbent in 50 mL of 0.1 M HCl. Stir 30 min. Filter and collect the acid filtrate. Measure lead, copper, and zinc.

Stage B: wash and dry (about 1.5 h). Wash three times with 50 mL DI. Dry at 50 °C for about 1 h.

Stage C: re-adsorption (120 min). Place the regenerated sorbent in fresh 50 mL of 50 ppm lead at the optimum pH. Stir 120 min. Filter and measure residual lead.

Calculate desorption efficiency $D(\%) = (\text{mass desorbed}/\text{mass adsorbed}) \times 100$; capacity retention

$R(\%) = q_e(\text{cycle } n)/q_e(\text{cycle } 1) \times 100$; and cumulative recovery from the pooled desorbates.

Expected. Capacity retention in the range 50–99% (target above 70%) after three cycles, with the desorbate showing acid recovery of the captured metals. Run all replicates of a cycle together before moving to the next.

9 Phase C: fixed-bed column

Column A runs three single-metal lead cycles for the retention and enrichment measurements; Column B runs one ternary breakthrough for multi-metal handling. Dimensions and conditions are from Section 5.4.

9.1 Packing and bed characterization

1. Place a 1 cm plug of acid-washed glass wool above the frit at the base.
2. Wet-pack 4.0 g of TA–OSS (0.50–1.00 mm) as a DI-water slurry, adding in increments and tapping gently to settle. Avoid trapped air and channels. Top with a 0.5 cm glass-wool plug and seat the adjustable adapter directly on the bed (no headspace).
3. Measure the settled bed height. Compute $V_{\text{bed}} = A \times \text{height}$ ($A = 0.785$ cm^2); cross-check by water displacement where feasible.
4. Compute $\rho_b = m/V_{\text{bed}}$ and $H{:}D$. Define BV $= V_{\text{bed}}$ in mL. Record ρ_b, V_{bed}, bed height, and $H{:}D$.

Expected. $\rho_b \approx 0.45$–$0.55\ \text{g cm}^{-3}$, bed height $\approx$ 9–11 cm, $H{:}D \approx$ 9:1 to 11:1.

Rationale. Wet-packing with tapping prevents air pockets and channeling; seating the adapter on the bed removes a top void that would broaden the front.

9.2 Pre-conditioning

1. Pump 5 BV of DI water at the optimum pH ($\approx$ 5.0) downflow at 8 BV h^{-1}.
2. Confirm the effluent pH stabilizes at the inlet pH (within ± 0.3) before the service run.

9.3 Service run and breakthrough (Column A, cycle 1)

Feed: 100 mg L^{-1} lead at the optimum pH, single metal.

1. Start the feed pump at 8 BV h^{-1} downflow (about 1.0 mL min^{-1} for an $\approx$ 8 mL bed; set from the measured BV).
2. Collect effluent fractions every 0.5 BV for the first 10 BV, then every 2 BV until $C/C_0 \approx 0.5$, then every 5 BV until $C/C_0 \geq 0.90$. Record the volume and cumulative BV of each fraction.
3. Measure each fraction for residual lead by AAS, prioritizing the breakthrough region and archiving the rest. Optionally record outlet pH.

Metrics: BV$_{10}$, BV$_{50}$, BV$_{90}$ from the curve; dynamic column capacity $q_{\text{col}} = [\int (C_0 - C)\, dV]/m$ by numerical integration.

Expected. A sigmoidal curve with effluent lead well below the feed before BV$_{10}$, confirming preferential adsorption and a reduced-lead effluent. BV$_{10}$ is a measured output, of order one hundred bed volumes at 100 mg L^{-1} and typically below the saturation estimate because dynamic capacity is 60–90% of equilibrium capacity.

9.4 In-column regeneration and lead enrichment

1. Stop the feed pump and drain the column to just above the bed.
2. Pump about 3 BV of 0.1 M HCl counter-current (upflow) at 4 $\mathrm{BV\,h^{-1}}$ (about 45 min). Collect the regenerate as a single pooled sample, or in 0.5 BV fractions if time permits.
3. Measure lead in the pooled regenerate (also copper and zinc for the Column B run).

Calculate $D(\%)$ = (mass desorbed/mass adsorbed in the prior service run) $\times$ 100 and the enrichment factor $\mathrm{EF} = C_{\mathrm{Pb,regenerate}}/C_{\mathrm{Pb,feed}}$.

Expected. Regenerate lead 10×–100× the feed (of order 50× at 100 $\mathrm{mg\,L^{-1}}$). Counter-current elution concentrates the desorbed lead into a small eluent volume.

9.5 Rinse

1. Rinse with DI water ($\approx$ pH 5) downflow at 8 $\mathrm{BV\,h^{-1}}$ until the outlet pH returns to within ± 0.3 of the service pH (typically 3–5 BV).

9.6 Cycles 2 and 3 (capacity retention)

Repeat Sections 9.3–9.4 for cycles 2 and 3 with the same feed, flow, and regeneration. Plot all three curves together; the per-cycle BV_{10} is the retention metric.

Note. Cycles 2 and 3 may be terminated once the effluent reaches $C/C_0 \approx 0.30$, because BV_{10} is captured before that point.

Calculate $R(\%) = \mathrm{BV}_{10}(\text{cycle } n)/\mathrm{BV}_{10}(\text{cycle } 1) \times 100$.

Expected. Retention at cycle 3 in the range 50–99% (target above 70%).

9.7 Column B: multi-metal (ternary) breakthrough

Demonstrates that the bed handles a multi-metal feed and preferentially retains lead, with a lead-enriched regenerate.

1. Pack a second column identically (4.0 g TA–OSS; characterize as in Section 9.1); pre-condition.
2. Feed 50 $\mathrm{mg\,L^{-1}}$ lead + 50 $\mathrm{mg\,L^{-1}}$ copper + 50 $\mathrm{mg\,L^{-1}}$ zinc at the optimum pH, downflow at 8 $\mathrm{BV\,h^{-1}}$.
3. Collect fractions (every 0.5 BV for the first 10 BV, then every 2 BV) and measure all three metals in a representative subset.
4. Continue until lead reaches $C/C_0 \geq 0.90$. Copper and zinc break through earlier; record any transient overshoot ($C/C_0 > 1$) as they are displaced by lead.
5. Regenerate once (counter-current 0.1 M HCl, about 3 BV at 4 $\mathrm{BV\,h^{-1}}$) and measure lead, copper, and zinc in the pooled regenerate.

Metrics: per-metal BV_{10} (expect copper and zinc below lead); lead fraction of the recovered regenerate.

Expected. Lead breaks through last and dominates the regenerate. The overshoot of copper and zinc is the signature of competitive displacement by lead. This run is analysis-heavy; it may be skipped if material or time is short (Section 8.4 already establishes selectivity) but is recommended.

9.8 End-state characterization

1. After the final cycle, unpack the column. Dry a portion of the spent TA–OSS at 50 °C and record its FTIR spectrum (the third FTIR sample of Section 7.5).
2. Photograph the column at packing, mid-breakthrough, and post-regeneration; photograph the spent sorbent.

Expected. The spent-sorbent FTIR shows shifts of the carbonyl/ester features and dampening of the phenolic O–H band relative to fresh TA–OSS, consistent with lead binding at the galloyl sites.

10 Sampling, calibration, and analysis by AAS

10.1 Calibration

- Five standards per metal (1, 5, 10, 25, 50 ppm) with 0.1 mL of 1 M HNO_3 per 100 mL, plus a blank. Each curve must reach $R^2 > 0.99$ before any sample is run.
- Run a mid-range check standard at the start, middle, and end of each batch.
- Samples above the top standard (100–300 ppm isotherm points, the 100 $\mathrm{mg\,L^{-1}}$ column feed, and the acid regenerates, which can reach thousands of ppm) must be diluted into range; record every dilution factor.

10.2 Sample handling

- Filter every liquid sample through 0.45 μm before analysis.
- If a sample is not run the same day, preserve it with a trace of 1 M HNO_3 and refrigerate.
- Label every sample with its phase, flask or fraction, nominal concentration, and dilution factor.

10.3 Instrument and quality control

- Flame AAS, air–acetylene, lines Pb 283.3 nm, Cu 324.8 nm, Zn 213.9 nm.
- One element per aspiration: ternary and Column B samples are read three times, once per lamp.
- Run a blank and a check standard with each batch; report replicated values as mean ± SD.
- Reject and re-run any batch where the check standard deviates by more than ±5–10% or a curve fails $R^2 > 0.99$.

11 Calculations and equations

C_0, C_e, C_t are initial, equilibrium, and time-point concentrations ($\mathrm{mg\,L^{-1}}$); V is volume (L); m is sorbent mass (g).

Quantity	Equation
Removal efficiency	$\text{Removal}\,(\%) = \dfrac{C_0 - C_e}{C_0} \times 100$
Equilibrium capacity	$q_e = \dfrac{(C_0 - C_e)\,V}{m}$
Time-point capacity	$q_t = \dfrac{(C_0 - C_t)\,V}{m}$
Langmuir (linear)	$\dfrac{C_e}{q_e} = \dfrac{1}{q_{\max} K_L} + \dfrac{C_e}{q_{\max}}$

Quantity	Equation
Freundlich (linear)	$\log q_e = \log K_F + \frac{1}{n}\log C_e$
Pseudo-first-order	$q_t = q_e\,(1 - e^{-k_1 t})$
Pseudo-second-order	$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{t}{q_e}$
Weber–Morris	$q_t = k_{\mathrm{id}}\,t^{1/2} + C$
Selectivity coefficient	$\alpha(\mathrm{Pb}/M) = \frac{q_{\mathrm{Pb}}\,C_{e,M}}{q_M\,C_{e,\mathrm{Pb}}}$
Distribution coefficient	$K_d = q_e/C_e$
Separation factor	$\beta_{i/j} = K_{d,i}/K_{d,j}$
Desorption efficiency	$D\,(\%) = \frac{\text{mass desorbed}}{\text{mass adsorbed}} \times 100$
Capacity retention (batch)	$R\,(\%) = \frac{q_e(\text{cycle } n)}{q_e(\text{cycle } 1)} \times 100$
Tannin loading	$(\%) = \frac{m_{\mathrm{func}} - m_{\mathrm{raw,eq}}}{m_{\mathrm{raw,eq}}} \times 100$
Residual mineral (ash)	$(\%) = \frac{m_{\mathrm{ash}}}{m_{\mathrm{dry}}} \times 100$
Packed-bed density	$\rho_b = m/V_{\mathrm{bed}}$
Aspect ratio	$H{:}D$ = bed height/column ID
Empty-bed contact time	$\mathrm{EBCT} = V_{\mathrm{bed}}/Q = 1/(\text{BV per hour})$
Breakthrough bed volume	BV_x: bed volumes when $C/C_0 = x$ ($x = 0.10, 0.50, 0.90$)
Dynamic column capacity	$q_{\mathrm{col}} = \frac{1}{m}\int_0^{V_{\mathrm{exh}}} (C_0 - C)\,dV$
Column capacity retention	$R_{\mathrm{col}}\,(\%) = \frac{\mathrm{BV}_{10}(\text{cycle } n)}{\mathrm{BV}_{10}(\text{cycle } 1)} \times 100$
Lead enrichment factor	$\mathrm{EF} = C_{\mathrm{Pb,regenerate}}/C_{\mathrm{Pb,feed}}$

12 Data recording and analysis

12.1 Data to record

- Calibration curves with R^2 values and blank readings (each batch).
- Initial and equilibrium concentrations (C_0, C_e) for every batch flask, with dilution factors.
- Time-point concentrations (C_t) for the lead and copper kinetic series.
- Column effluent-fraction concentrations against cumulative bed volume, for Column A (three cycles) and Column B.
- Regenerate concentrations (pooled or fractionated) for every regeneration.
- Bed characterization: ρ_b, V_{bed}, bed height, $H{:}D$.
- Ash results and tannin-loading masses.
- FTIR spectra (RAW–OSS, TA–OSS, spent) and a labelled overlay; $\mathrm{pH_{PZC}}$ data; photographs.

12.2 Analysis to perform

Langmuir and Freundlich fits and $q_{\max}$; pseudo-first-order, pseudo-second-order, and Weber–Morris fits; selectivity coefficients and separation factors; breakthrough curves with BV_{10}, BV_{50}, and BV_{90}, plus q_{col}; desorption efficiency, capacity retention, and enrichment factor.

12.3 Record-keeping

Keep a bound or validated electronic notebook; label every flask, vial, and fraction; log any deviation with its reason at the time it occurs; keep instrument and calibration records with the dataset.

13 Run and analysis-count summary

Approximate counts for planning. Column fractions assume adaptive sampling (below).

Experiment	**Flasks / runs**	**AAS determinations**
pH optimization (4 TA, 4 RAW, 1 control)	9	≈ 10 (lead)
Single-metal isotherms (lead 6 incl. n=3; copper 4; zinc 4; RAW)	17	≈ 16
Kinetics (lead 7, copper 7)	14	≈ 14
Competitive ternary (equimolar n=3, RAW; minority n=3; initial aliquots)	7	≈ 27 (3 per sample)
Batch regeneration (3 cycles × n=3: desorb and re-adsorb)	18 runs	≈ 36
Ash / residual mineral (duplicate each)	4	0 (gravimetric)
Point of zero charge (9 TA, 9 RAW)	18	0 (pH only)
FTIR spectra (RAW, TA, spent)	3 spectra	—
Column A, three cycles (fractions, subset; 3 regenerates)	≈ 190 fractions	≈ 100 (lead)
Column B ternary (fractions, subset; 1 regenerate)	≈ 80 fractions	≈ 123 (3 per sample)
Indicative total	—	≈ 330 (up to ≈ 450)

Note. Collect column fractions on the fine/coarse schedule of Sections 9.3 and the ternary run, but analyze the breakthrough-region fractions first and archive plateau fractions; run the archived ones only if the curve needs refining.

Note. Material check: TA–OSS demand is about 4.9 g of batch flasks, 8 g for the two beds, 1 g for ash, and a little for FTIR, against roughly 18–19 g produced from the 18 g charge. RAW–OSS demand is about 2.6 g of the 5 g portioned. Both close within the ≈ 25 g of supplied ossein.

14 Suggested execution sequence

Indicative; actual durations depend on the measured breakthrough length. Run calibration in parallel from day 1.

Phase / day	**Activity**
A, day 1	Rinse, dry, grind, sieve the ossein; start functionalization and the RAW control on the stirrer (overnight); begin calibration

Phase / day	Activity
A, day 2	Wash and dry both batches; determine tannin loading; portion; start FTIR and ash
A, day 3	Finish FTIR and ash; run point of zero charge
B, day 4	pH optimization; begin single-metal isotherms
B, day 5	Finish isotherms; run lead and copper kinetics; interleave analysis
B, day 6	Competitive ternary (equimolar and minority); begin batch regeneration
B, day 7	Finish batch regeneration; complete Phase B analysis
C, day 8	Pack and characterize Column A; pre-condition; begin cycle-1 service run
C, days 9–10	Complete cycle-1 breakthrough (overnight standby allowed); regenerate and rinse
C, days 11–14	Cycles 2 and 3 (may terminate at $C/C_0 \approx 0.30$); regenerate and rinse each
C, days 15–17	Pack Column B; ternary breakthrough; single regeneration
C, days 17–18	Unpack; spent-sorbent FTIR; finalize photographs and dataset

Note. A 200 $\mathrm{mg\,L^{-1}}$ feed and/or a velocity up to about 15 $\mathrm{BV\,h^{-1}}$ roughly halves each service run and can shorten Phase C by several days, at the cost of a broader mass-transfer zone.

15 Appendix: data-recording templates

A. Batch observation log

Date	Expt	Sorbent / mass	Metal / C_0	pH i/f	Vol/time	Filtrate	Notes

B. Column-run datasheet

Header (one per run): column ID; sorbent mass (g); bed height (cm); V_{bed} / BV (mL); ρ_b ($\mathrm{g\,cm^{-3}}$); H:D; feed C_0 ($\mathrm{mg\,L^{-1}}$); pH; flow ($\mathrm{BV\,h^{-1}}$ and $\mathrm{mL\,min^{-1}}$); EBCT (min); date.

Frac	Cum. vol (mL)	Cum. BV	Time	Effluent Pb (Cu, Zn)	C/C_0	Outlet pH	Notes

Regeneration: eluent and molarity; direction (counter-current); flow ($BV\,h^{-1}$); volume (BV); pooled regenerate Pb (Cu, Zn); enrichment factor; desorption efficiency.

Eluent / M	Direction	Flow	Volume	Regenerate Pb (Cu, Zn)	EF / D

Rinse: bed volumes of DI to restore outlet pH to within ±0.3 of inlet; final outlet pH.

Rinse (BV)	Inlet pH	Final outlet pH	Notes